Jayashwanth Jeganathan Subhashini

Selected Engineering Projects

Embedded Systems | IoT | Experimental Systems | R&D

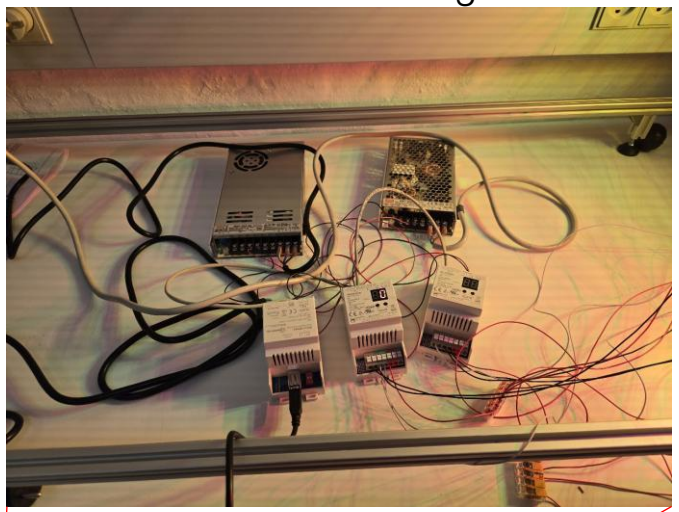
M.Sc. Electrical Engineering, DTU

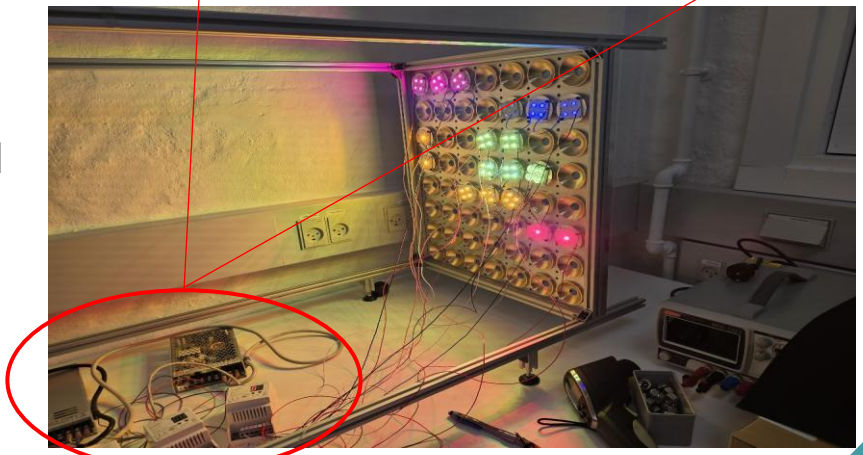
Research Assistant | DTU Physics – Solar Simulator

Solar Simulator — Hardware Development & Testing

- Led electrical and system development of a high-intensity LED solar simulator, from prototyping through commissioning and validation.
- Designed PCBs and integrated power supplies, electronic hardware, and commercial Zencontrol IoT lighting control equipment.
- Built and wired experimental setups to test LED panels, power supplies, and control equipment.
- Performed hardware bring-up, electrical measurements, thermal characterization, and optical-power testing.
- Used laboratory instrumentation to investigate system behaviour, troubleshoot issues, and verify performance.
- Documented experimental setups, measurement results, and validation procedures.
- Developed a system producing approximately **60 times the intensity of natural sunlight**.

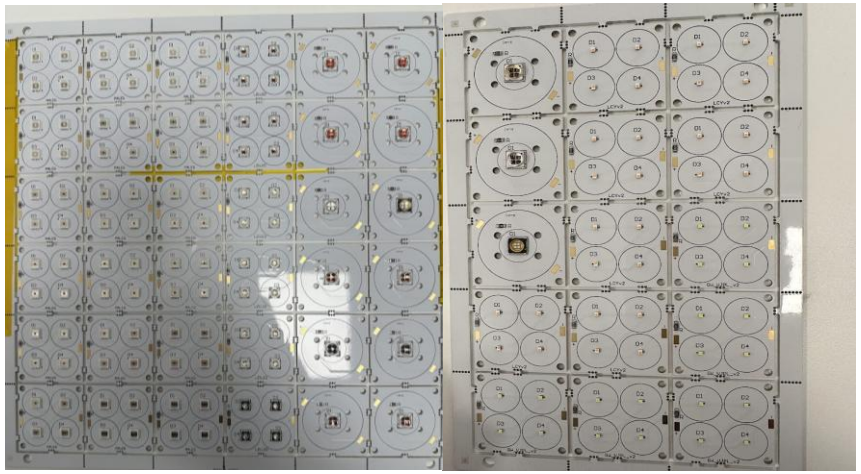
Power Supply and Dimmer Setup for LED Testing



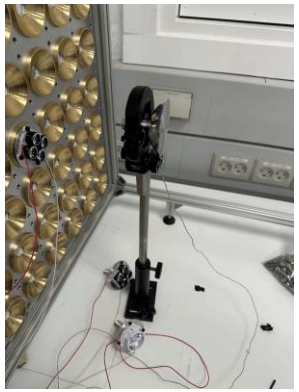


Prototype Setup for Optical Power Measurement at Different Distances

LED PCBs Developed for the Simulator



Test Setup for Thermal Characterization



Research Assistant | DTU Physics

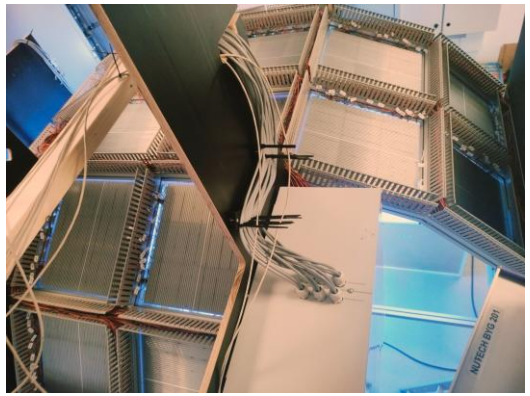
System Integration, Troubleshooting & Validation

- Assembled and wired dedicated controller, dimmer, and power-supply enclosures.
- Built a **2.0 × 1.6 × 0.4 m enclosure** housing approximately **80 power supplies**, with a combined supply current requirement of **80 A**.
- Integrated the electrical, lighting control, power, and optical subsystems into a complete experimental platform.
- Performed electrical, thermal, optical-power, and spectral measurements to evaluate system performance.
- Investigated hardware and system-level issues during integration and commissioning.
- Prepared electrical schematics, test documentation, and measurement records.

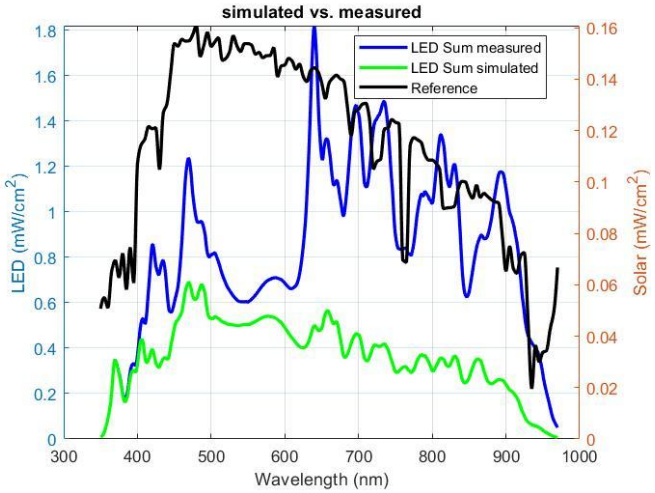
Outcome: Delivered a validated, high-intensity solar simulator for bleaching rock samples used in luminescence-dating research.



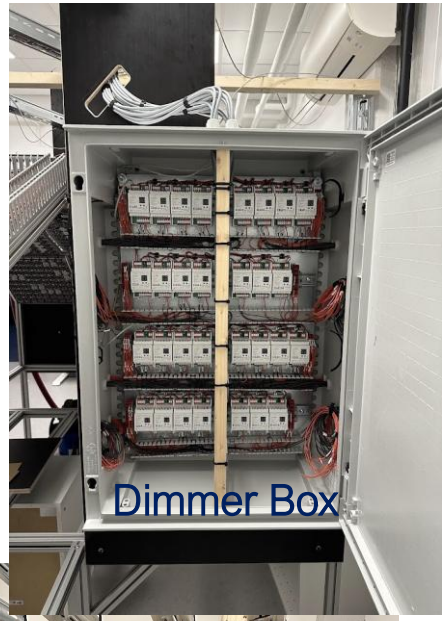
Final System During Operation – Concentrated Illumination at the Sample Position

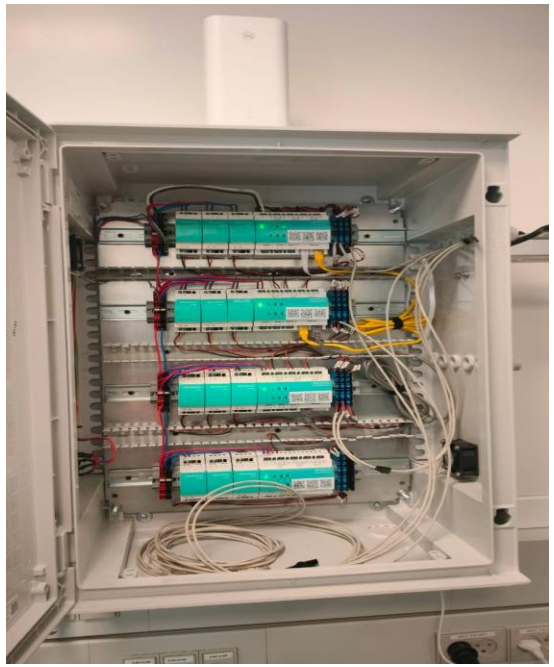


Final System in Lab



Spectral Comparison for One Panel – Simulated vs. Measured Spectrum





Controller Box



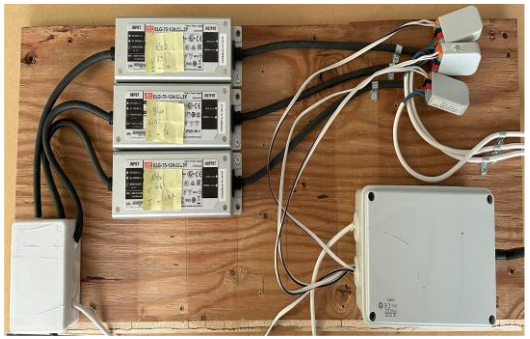
Power Supply Enclosure – 80 A

Thesis Student & Junior Embedded Systems Developer | LEDiBond A/S

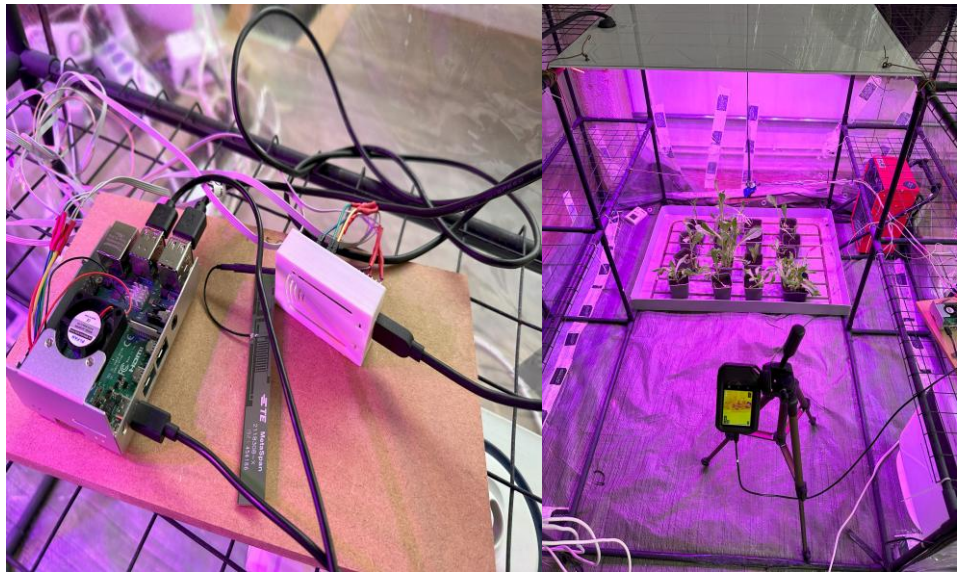
Environmental Sensor Monitoring & IoT Connectivity

- Developed and tested ESP32- and Raspberry Pi-based embedded prototypes for environmental sensor monitoring.
- Developed C firmware for sensor acquisition and communication.
- Developed and executed **Python scripts on a Linux-based Raspberry Pi** for Azure IoT connectivity and sensor-data transfer.
- Integrated environmental sensors and MQTT-based communication between embedded devices.
- Performed sensor calibration, hardware testing, communication debugging, and measurement validation.
- Improved prototype functionality through hardware–software integration and iterative testing.

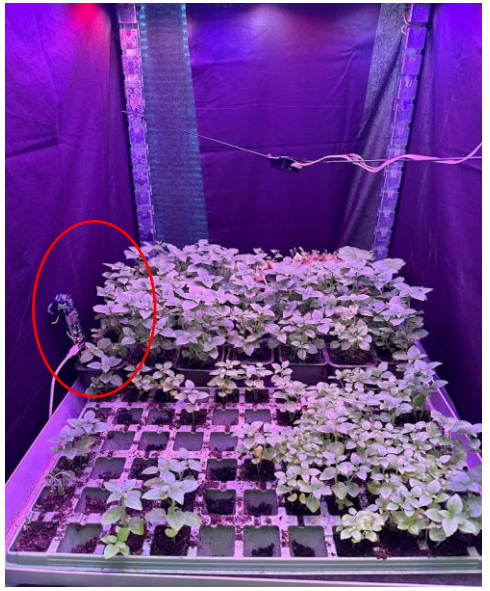
Outcome: Developed and validated a connected sensor-monitoring prototype integrating embedded hardware, firmware, environmental sensors, and cloud IoT connectivity.



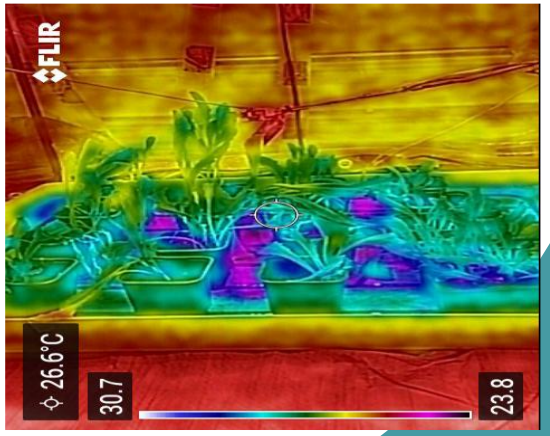
Electrical Setup for System Testing



Plant Test Setup with Sensor Data Acquisition System



Second Prototype Developed Toward a Commercial Product



Leaf Temperature Measurement Using a Thermal Camera

Student Assistant | DTU Electro

Growbot — Embedded Device Assembly & Testing

- Developed, assembled, tested, and delivered **25 Raspberry Pi- and Arduino-based embedded systems**.
- Integrated sensors, actuators, LEDs, pumps, motors, and electronic control hardware.
- Performed PCB assembly, soldering, wiring, hardware integration, and bring-up.
- Carried out functional testing, troubleshooting, and fault diagnosis during development.
- Implemented hardware improvements based on testing and validation results.
- Integrated sensor measurements with a web interface for monitoring and data storage.
- Collaborated with researchers, engineers, and an external development partner.

Outcome: Delivered 25 tested educational embedded systems to schools in Denmark.



Project Closing Conference with Researchers and Teachers



System Assembly, Testing & Troubleshooting



Final Systems Before Deployment to Schools